NOVA SCHOOL OF SCIENCE & TECHNOLOGY

**Interação Pessoa-Máquina**

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# Music Maker Hub

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## Individual Assignment: Lab class 1


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## Good User Interface Design

The objective of a tilt-and-turn window is to provide two different ways of opening a window for users using a single handle.

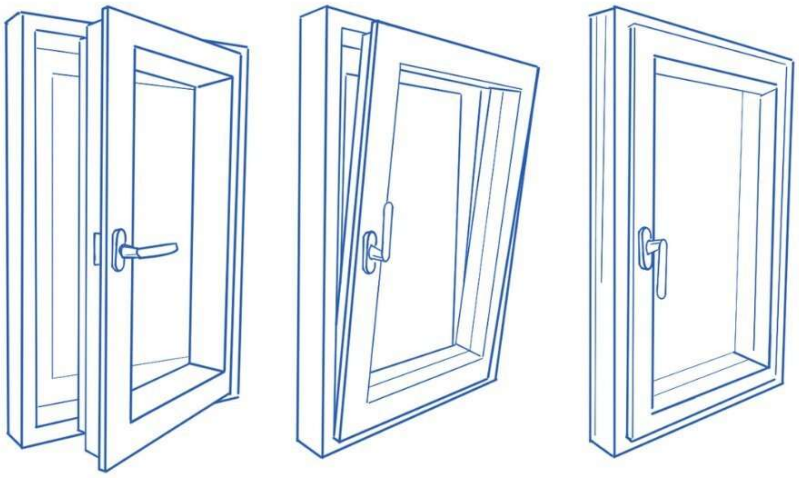
- Tilt: Upper part of the window opens inwards, allowing a continuous ventilation gap without fully opening the window
- Turn: Side of the window opens inward, allowing full opening and easy access for cleaning

Good aspects:

- Users can control the ventilation intensity precisely
- Users can use the window easily with one hand
- Users can clean easily clean the window from the back
- Users can close the window tightly against the frame (which prevents noise and unwanted air exchange)

Bad aspects:

- Users have higher complexity in handling the handle.
- Users have a risk of incorrect operation
- Users need space inside the room to open the window



Explanation: I think this design is a good example of user interface design, because it gives a lot of control for the ventilation, is very easy to use with one handle, is easy to clean from outside and it protects well against noise and is well sealed against heat and cold from outside. In addition, it is very uniformly designed and additional features like handle key locks can be implemented to make it safe for toddlers.

The bad aspects are the higher complexity for first-time users and connected incorrect operation. As an improvement the handle can do a click sound when it is on a supposed position and the general difficulty to take the window off its hinges can be adjusted. For the space inside the room, the general position of the windows can be adjusted, for example in class rooms it should be over the heads of seated students.

## Bad User Interface Design

The objective of a sliding window is to allow users to open and close a window by moving one or more glass panels horizontally along the frame.

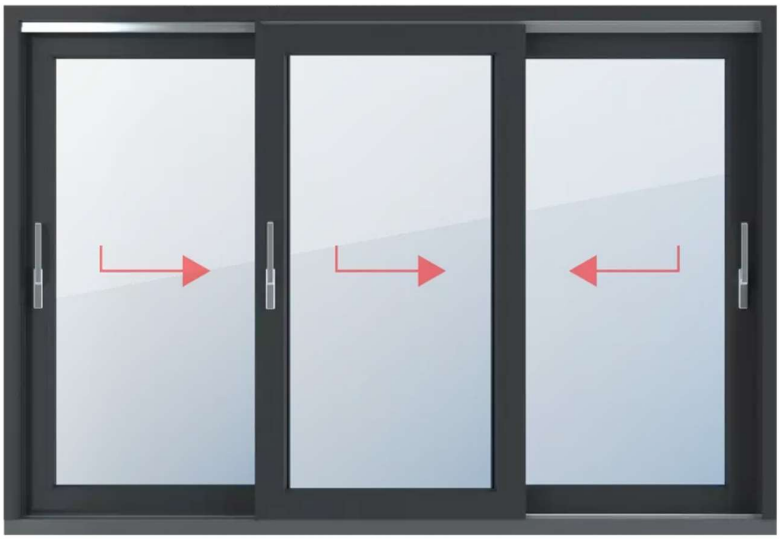
- Sliding: Users moves panel sideways along fixed track, only parts of the window can be opened at a time

Good aspects:

- Users can slide the window to different positions and adjust to ventilation needs
- Users slide window sideways instead of opening it into the room
- Users can understand the basic interaction of sliding easily

Bad aspects:

- Users can commonly only open half of the window and maximum ventilation is limited
- Users have missing feedback about the locking state as the locking mechanism is difficult to recognize
- Users have difficulties in operation due to friction with the sliding tracks and dirt inside
- Users cannot easily clean the window from outside
- Users need to use inconsistent and different handles and lock systems
- Users have difficulties in sealing the window tightly (more noise and unwanted air exchange)



Explanation: I think this design is bad, because is not possible to fully open the window, users miss uniformity in design, cleaning is more difficult and operation is difficulty because of friction. Additionally, it doesn´t protect against noise and is sealed less.

It can be improved with more uniform handles and locks. Also, the friction can be reduced with well designed sliding tracks. For the unwanted air exchange and noise protection the sealing can be improved.

Generally, in my explanation I am influenced by a user bias, because I am only used to the (in my opinion) good user interface window. In general, the requirements for windows are different in Germany and Portugal, for example there is a different emphasize on protection against cold and heat.